

LatticeAnalytics: Strut-Level Visualization and Inspection of Additively Manufactured Lattice Structures

Haichao Miao , Saurabh Narain , Vuthea Chheang , Garrett Hooten, Raiyan Seede, Pavol Klacansky, Kaila Morgen Bertsch, Gabe Guss, Brian Giera , and Peer-Timo Bremer 

Abstract—Additive manufacturing (AM) is revolutionizing the production of custom components with complex internal geometries, essential for high-performance applications in diverse fields such as medicine and defense. These AM parts optimize strength while minimizing weight by utilizing internal lattice structures consisting of large quantities of small interconnected struts. However, the complexity of these structures, combined with the challenges of using X-ray Computed Tomography (XCT) data, makes validation of part reliability difficult. This ultimately inhibits the development of novel parts for our collaborating material scientists. We introduce *LatticeAnalytics*, a novel framework specifically designed for visual inspection of defects in these lattice structures. Our framework offers an end-to-end solution that includes the data management of XCT scans, enables remote access for geographically dispersed teams through a web-based dashboard, and incorporates novel visualizations. Our analysis is facilitated by a coarse alignment between the lattice's nominal model, a spatial graph, and the XCT data. We employ a simple VR-based approach for fast and rough alignment, followed by an offline registration and identification of the struts. With the nodes and struts aligned and identified in the volume, our framework allows querying of subvolumes containing a single strut at multiple resolutions. This avoids computation over the entire lattice and also allow for easy parallelization of down-stream computations, such as strut-specific metrics. To depict a fast overview of the strut quality, we introduce two innovative visual encodings, crucial for our collaborators' research in creating novel AM parts: the *Contour View* and the

Roughness Map, which depict critical geometrical and surface features of individual struts in standardized two 2D views. We evaluated the integrated system through expert interviews. The feedback confirms the framework's practicality and its effectiveness in enhancing current inspection workflows. It solves major bottlenecks for our collaborators, ultimately helping them create novel parts with advanced properties.

Index Terms—Additive manufacturing (AM), 3D printing, defect analysis, visual abstraction, projection, mapping, registration, virtual reality, dashboard, lattices, struts, x-ray computed tomography (XCT), non-destructive evaluation.

I. INTRODUCTION

ADDITIVE Manufacturing (AM) is revolutionizing industry sectors, such as medicine, aerospace, and defense, by enabling on-demand fabrication of custom parts with unprecedented material properties, e.g., light weighting without loss of strength, via previously unrealizable designs [1], [2]. Their unprecedented capabilities are largely driven by additive techniques such as Laser Powder Bed Fusion (LPBF) [3] which creates intricate lattice structures comprised of repeatedly arranged cylindrical elements called struts. With advancements, current AM processes face significant challenges associated with unpredictable and/or unavoidable defects that may affect part performance. Common defects include unwanted strut deformations (bending or breakage), and irregular surface roughness all of which can impact the mechanical behavior of the part. Currently, there is no uniformly adopted analysis approach, even after decades of research. Thus, the manual inspection by a human expert remains crucial, especially in a research and development environment where part specifications and geometry change quickly. Given the high manufacturing costs, and variability in production, X-ray Computed Tomography (XCT) stands out as the gold standard for non-destructive analysis of internal material densities. XCT scans of AM parts often result in complex and large datasets, exceeding $4k^3$ voxels, which can be difficult to access and analyze due to their size. Moreover, the design, fabrication, and inspection are typically carried out by a diverse team of non-destructive evaluation experts, material scientists, and research technicians that are often geographically dispersed. Hence, the inspection cost exceeds the fabrication cost by a factor of four according to our collaborators. The problem is that bottlenecks exist at several stages in the

Received 20 November 2024; revised 10 July 2025; accepted 13 July 2025. Date of publication 6 August 2025; date of current version 5 September 2025. This work was supported in part by the U.S. Department of Energy by Lawrence Livermore National Laboratory under Grant DE-AC52-07NA27344, and in part by the LLNL LDRD under Grant 23-SI-003 and Grant LLNL-JRNL-871130. Recommended for acceptance by V. Natarajan. (Corresponding author: Brian Giera.)

This work involved human subjects or animals in its research. Approval of all ethical and experimental procedures and protocols was granted by Lawrence Livermore National Laboratory (LLNL) review committee under Application No. LLNL-JRNL-871130.

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This article has supplementary downloadable material available at <https://doi.org/10.1109/TVCG.2025.3593230>, provided by the authors.

Digital Object Identifier 10.1109/TVCG.2025.3593230

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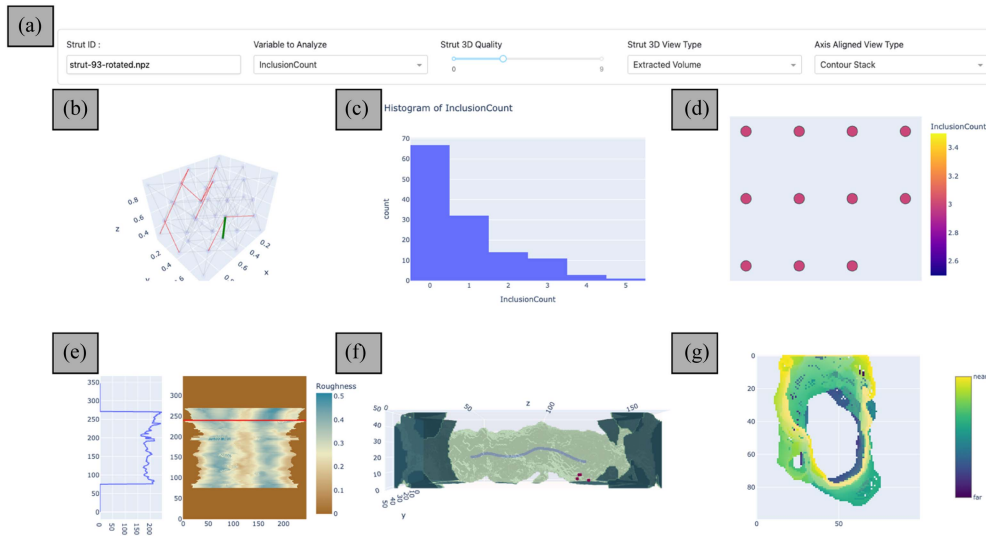


Fig. 1. The *LatticeAnalytics* dashboard, created for material scientists studying lattice structure defects. (A)Control Menu, (B)Spatial Graph, (C)Metric Histogram, (D)Strut Overview, (E)Roughness Map (RM), (F)Multi-resolution iso-surface rendering, and (G)Contour View (CV).

workflow, including challenges in data management, segmentation, registration, and analysis.

To address critical issues in inspection, we developed a new defect analysis framework in collaboration with domain experts that provides a comprehensive solution for the semi-automated inspection of these intricate parts. Since our project was embedded in an LPBF context, we demonstrate our approach on lattice structures produced by LPBF. However, our technique is generalizable to any lattice-based AM technique that uses XCT for inspection. Unlike existing work that often focus on a single step of the process, our *LatticeAnalytics* framework offers a generalizable, end-to-end solution that overcomes critical issues in the current workflow, while providing APIs to combine with existing approaches. The key idea is that we align the nominal spatial graph of a lattice structure with the corresponding XCT data to identify the nodes and struts. Unlike recent approaches [4], [5], [6], [7], we focus on the human in the loop and effectively combine the expert's intuition and experience with scalable automated algorithms that supports a novel defect inspection workflow. Specifically, our key contributions include:

- *Domain characterization and collaboration*: We characterized the problem domain through long-term collaboration with materials scientists and NDE experts, which revealed several visualization, data management, and processing challenges. We addressed these challenges in our framework, and we evaluated our solutions with domain experts to confirm the benefits, usability, and applicability.
- *Comprehensive pipeline for lattice analysis*: Our semi-automated approach utilizes manual user intervention in steps only where necessary (during the alignment stage and result interpretation) during the alignment and the interpretation of results. This includes a VR-based alignment that roughly registers a part's spatial graph with the actual CT volume, going beyond existing approaches based on surface to surface/volume correspondence. With this crucial alignment, our pipeline automatically extracts key

metrics designed to be parallelized, and provides APIs that allows users to customize metric computations.

- *Tailored dashboard and visual encoding*: We developed a dashboard to allow the disparate team to remotely access the large-scale XCT volumes in real-time and interpret the results, facilitating a faster and more responsive workflow. At the heart of the dashboard, we integrated two visualization techniques – the *Roughness Map (RM)* and the *Contour View (CV)* – that succinctly represent strut geometry and its surface roughness in 2D views respectively.

II. RELATED WORK

Our work is related to previous approaches in XCT analysis that are focused on the additive manufacturing domain. We also investigate practical solutions to detecting defects as well as the visualization of objects and defects in large-scale volumes.

A. XCT Analysis and Visual Characterization in AM

Inspection and analysis of AM lattice structures can be classified into four categories: internal defects, external defects, accuracy of dimension, and surface roughness [8]. Most inspection analyses often rely on techniques, such as ultrasonic testing [9], infrared [10], [11], and XCT [12], [13], to analyze and create evaluation criteria for quality control, processing techniques, and optimization. XCT offers a nondestructive evaluation of the internal features, and does not suffer from the inability to detect steep slopes and undercuts [14].

Visual characterization and interactive visualization techniques, including visual computing are crucial in material science to provide insights into complex material properties [6], [15]. Klingaa et al. [4] presented an approach using XCT and image analysis to extract and quantitatively characterize the roughness on internal surface made by metal AM. Their results indicated that internal roughness cannot be captured by a single global measurement, because roughness analysis

for metal AM requires information from multiple orientations along the surface. Zeltmann et al. [16] demonstrated defect detection scenarios and challenges in the AM process chain with embedded defects and effects of altered printing direction using non-destructive methods, including XCT and ultrasonic imaging. Similarly, Okarma et al. [5] discussed techniques to improve visual quality assessment of color surface for AM based on image entropy. This is important because surface quality may correlate with physical or mechanical properties, which is helpful for quality monitoring. Furthermore, Yang et al. [7] investigated the effect of a density gradient on fatigue behaviors and properties of gyroid graded lattice structures produced by LPBF. They highlighted the benefits of the gradient design in improving fatigue performance.

As previous studies have shown, XCT was often used for measurements and characterization as well as verification with engineering design for additive manufacturing. Identifying defects and their characterizations may require high-resolution XCT scans, which can be challenging to process because their large-size, complexities, and spatial characteristics [17]. In contrast, our work incorporates several visualization techniques, including spatial graph, metric histogram, roughness map, and multi-resolution volume rendering for large-scale XCT data, which provide a comprehensive analysis for lattice analysis and characterization.

B. Defect Visualization in Large-Scale Volumes

Defect visualization is crucial in additively manufactured lattice structures for several reasons ranging from early detection of the problems to quality control and future failure prevention [18]. The visualization helps the user to identify problems early in production, allowing for corrective actions and analysis, saving time, and minimizing resources. Additionally, understanding the types and distributions of defects within materials in AM provides valuable insights and can be crucial for material characterization in optimizing material design and development.

Local defects can occupy a small fraction of the entire volume. However, it needs appropriate approaches or techniques to visualize them when it comes to a large-scale volume, which requires resource-intensive computation and rendering [19], [20]. Tominski et al. [21] discussed virtual interactive lens techniques to provide alternative representations for selected regions of interest in large-scale visualization. Such techniques allow defect details to be visualized at full resolution within an interactive lens, while the rest of the volume is shown at low resolution. Mota et al. [22] presented a 3D interactive lens technique by combining 3D multi-geometry data and decal lenses for focus and context visualization. Similarly, Kluge et al. [23] presented interactive lenses as embodied tools for immersive analytics on volumetric data. While the use of interactive lenses allows the user to focus on a specific region of interest and load the subset of the dataset at full resolution, our proposed system allows the user to choose and visualize representative exemplars that retain a connection to the low-level

data, allowing the user to quickly validate and analyze the data.

Besides the advantages in depth perception in VR, natural interaction offers one key advantage for immersive alignment. A study conducted by Yang et al. [24] demonstrated a variety of approaches for navigating and exploring immersive data. They compared four conditions to enable analysts to navigate and explore data at different levels of scale: room-sized interface, room-sized interface with an overview, zooming interface, and zooming interface with an overview. The results suggest that if the user has the capability to perform the task in a large open space, the room-sized interface would benefit; however, the zooming interface is beneficial for tasks that require a significant effort in number-of-travels and context switching.

To our knowledge, the task of aligning spatial models has not thoroughly explored in the past, however, our choice of VR for alignment is inspired by several related works [25], [26], [27]. Using VR for alignment offers several benefits compared to desktop-based modalities, including enhanced spatial awareness, intuitive interaction and manipulation, reduced cognitive load, and increased user confidence and engagement [28], [29], [30]. A number of techniques have been developed for manipulating and aligning objects in immersive environments, including selection through progressive refinement [31], *World-in-Miniature* metaphor [32], using pins or control points attached to the object [33], and using widgets [25]. Our project is highly influenced by the work of Klacansky et al. [27] and Rodrigues et al. [26] who demonstrated the advantages of using VR for precise alignment, particularly between the the iso-surface of the XCT scan and their nominal CAD model. Instead of their surface-to-surface alignment method, which impose significant requirements in real-time rendering and interaction, our approach relies only on a spatial graph, which has a low memory footprint, and direct volume rendering. Furthermore, the key difference to the existing work lies in our data management approach that enables us to query only a small part of the volume, containing a single strut instead of loading the entire volume into memory for analysis. While VR is utilized for alignment between spatial graph and 3D volume at this stage, the actual analysis is facilitated on a desktop screen, which requires appropriate visual representations in 2D.

C. Visual Analytics and Defect Detection

A human-centered approach that relies on visual representations of features can provide a flexible workflow for experts. This approach helps them identify patterns and root causes of defects, especially in AM use cases. Since XCT volumes are often large, visual analytics should aim to process large-volume data quickly to provide the analysis and enhance the inspection process [34], [35]. Laha et al. [19] discussed significant benefits for analyzing volume data in immersive environments, including higher levels of immersion, which lead to more effective analysis of spatially complex structures. In addition to 3D volume representation, projection and unwrapping techniques are often used to transform complex 3D structures into simpler 2D representation, a

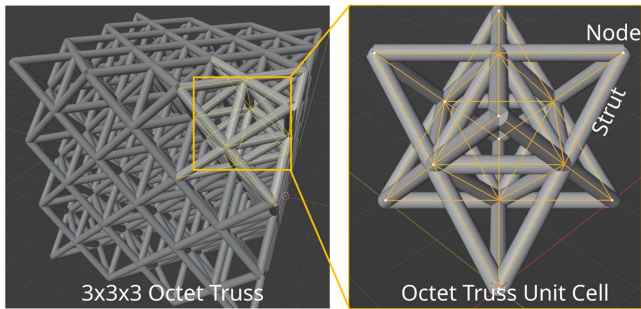


Fig. 2. The basic geometry of an octet truss structure. The unit cell of the octet truss is based on an octahedron, described by a spatial graph that consists of nodes and edges in 3D space.

traditional technique for visualization that makes it easier to gain an overview and interpret the data [36].

Skeleton-based analysis is another avenue to consider, given the tubular structure of struts. Skeletons simplify complex geometry and increase computational efficiency; accordingly. This is one of the reasons we extract the centerline of each strut as part of our analysis. Skeleton-based analysis for complex geometries provides an easier way to analyze and interpret the structural integrity and properties of manufacturing parts. El Jai et al. [37] introduced a benchmark for scanning strategy that can be adapted in the AM process. Their proposed strategy is also based on the skeleton of shapes, which shows potential in applying to more complex geometries and characterizing the properties of parts printed using the scanning pattern. Azman et al. [38] proposed a skeleton model to create and store lattice structures using points, lines, joints, and their sections for AM. They demonstrated that this skeleton model can be used to support efficient data representation and storage, reducing file size and creating conformal and non-conformal lattice structures.

Visual encoding approaches, including *Roughness Map* and *Contour View*, are essential for visualization and robust defect detection [39], [40]. While such approaches are common for visualizing geometric structures, in this work we introduce a bespoke strut-centric encoding designed specifically for strut-level visualization. This encoding comprises two components: (i) each strut is abstracted into a layered, color-coded 2D contour view that reveals its geometrical shape, and (ii) strut surface roughness is represented as a map utilizing a topographic metaphor. This per-strut representation fundamentally differs from traditional global iso-surfaces or sectional contours commonly cited in the literature. Our *Roughness Map* aligns broadly with flattening techniques [41] developed to visualize tubular structures in a 2D view. While these techniques have not yet been applied to strut-based AM structures, they have been widely explored in the medical domain for vascular visualization. These methods are based either on surface deformation [42] or forms of reformation [43], [44] to provide an unoccluded view of the tubular structures. Although we conceptually adopt these techniques, the key difference in our strut-based approach is our primary interest in visualizing protrusions and cavities on the strut's surface. This necessitates an additional feature extraction step to compute deviations from the nominal surface, which we

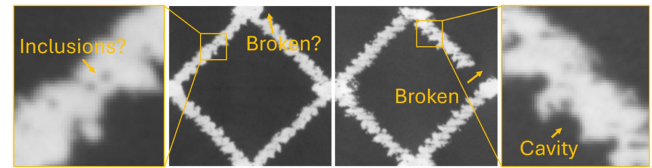


Fig. 3. A typical slice of a CT scan of a lattice structure with common defects, such as porosity (inclusions), surface roughness (cavity and material protrusions), and broken struts. A material scientist expert is typically required to manually determine the extent of the defect.

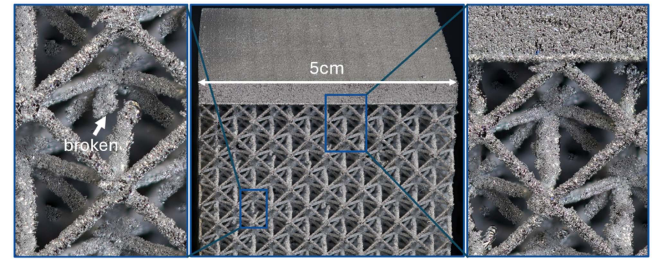


Fig. 4. A photo of a $7 \times 7 \times 7$ octet truss structure made from Ti64 alloy. A broken strut is visible from the outside, however, struts on the inside are occluded. At this close-up image the rough strut surface are clearly visible.

represent through color encoding. Our *Contour View* aims to abstract the geometry of the strut by projecting the cross-sectional shape along the longitudinal axis. Most research in this area, including visual encoding, often stems from medical domains, targeting the visualization of cross-sections of tubular anatomical structures [45]. The Contour View employs the concept of projecting data from n dimensions to $n - 1$ dimensions, similar to the underlying idea of the space-time cube [46]. Instead of projecting along the temporal axis, our Contour View projects along the longitudinal axis of the strut and also employs color coding to represent the position along this axis.

III. METAL ADDITIVE MANUFACTURING AND PROJECT BACKGROUND

This research is driven by strategic investments within a national laboratory to develop novel techniques for characterizing defects in AM parts produced via LPBF, an important method for advancing manufacturing capabilities crucial for defense and national security applications. LPBF operates by selectively melting and fusing layers of metal powder using a high-power laser beam. This beam selectively melts the powder to form the intended part layer by layer. One significant advantage of LPBF is its ability to fabricate complex metal parts with intricate internal geometries (example shown in Fig. 4), such as lattice structures composed of interconnected struts and nodes. These structures are often configured as repetitive unit cells, like the octet-truss (Fig. 2), which is known for their optimal strength-to-weight ratios and isotropic mechanical properties. Such designs not only reduce the overall weight but are also engineered to meet specific functional requirements of our applications.

Despite these innovations, the internal lattice structures can exhibit defects that potentially compromise part performance, safety, and durability. These defects manifest in several forms,

as illustrated in Fig. 3: Reduced structural integrity is affected by **bent and broken struts** that are caused by insufficient support during the print, residual stress, or material impurities. Reduced mechanical strength, due to **porosity** is caused by insufficient melting and trapped gases. Finally, reduced fatigue life associated with a **rough surface** of the struts is caused by partial melting. Given these challenges, the only feasible approach to ensure the reliability of these parts for their intended applications is through the use of advanced, non-destructive imaging techniques. Subsequent inspection by a human expert is still required. This project aims to enhance the understanding of these defects. It also seeks to develop methodologies to effectively identify and characterize them, thereby ensuring the robustness and reliability of AM components in critical applications.

IV. DATASET DESCRIPTION

Traditional approaches often align a nominal CAD mesh to XCT scan data [27]. In our approach, we instead utilize a spatial graph description of the nominal lattice topology, combined with high-resolution XCT scans of the printed parts. The *spatial graph* defines the lattice as a network of nodes and edges, with each edge representing a strut characterized by thickness properties corresponding to its radius and circular shape. This graph serves as the base design specification and, when combined with the CAD model and build parameters, accurately represents the nominal part geometry. Our system processes the nominal lattice data via custom JSON files (generated by an in-house tool), and it also supports the standard 3MF (3D Manufacturing Format) for lattice specifications. The printed parts, created via LPBF, are imaged using XCT to produce volumetric datasets. Due to the large size of volumetric data, we convert these datasets into the OpenViSUS multi-resolution format for efficient data management and visualization [47]. Using OpenVisus, we can effectively query down-sampled data or explore specific subspaces at full resolution and avoid unnecessary I/O operations, e.g., querying only a sub-region of the volume. The processed data and the derived metrics are organized into CSV files, containing strut-specific information. Our *LatticeAnalytics* dashboard provides access to the volume data, these CSV files, and the processing algorithms.

Our collaborators tested *LatticeAnalytics* on various parts (Table I) that are composed of octet trusses with varying unit cell sizes. The main dataset, the $2 \times 2 \times 1$ *AI Lattice*, is used to demonstrate our methodology in Section VI. As ground-truth is not available for real data, we also tested our approach on a public dataset we previously published.¹ The simulated CT scan contains an $8 \times 8 \times 8$ lattice with known defects. To showcase scalability to large-scale volumes, we also applied our framework to a $9 \times 9 \times 9$ *AMTruss* ($\sim 91GB$) and analyze results from ten $3 \times 3 \times 3$ truss structures that our domain experts are currently printing and conducting experiments on.

The pipeline processes each strut independently, giving **O(#struts)** complexity and parallel scalability. On typical GPU

TABLE I
OUR COLLABORATORS PROVIDED US WITH THESE DATASETS THAT DEPICT PARTS USED IN THEIR RESEARCH ON AM

Name	# Struts	Size (Unit Cells)	Resolution	Size (GB)
AI Lattice	128	$2 \times 2 \times 1$	$1072 \times 1395 \times 800$	5.26GB
AMTruss	18,468	$9 \times 9 \times 9$	$2900 \times 2900 \times 2600$	91.33GB
Five Defects	13,056	$8 \times 8 \times 8$	$1199 \times 1199 \times 1199$	6.43GB
2P10	756	$3 \times 3 \times 3$	$1000 \times 902 \times 1139$	1.63GB
5P6	756	$3 \times 3 \times 3$	$943 \times 857 \times 1099$	2.01GB
7P9	756	$3 \times 3 \times 3$	$985 \times 864 \times 1134$	2.10GB
8P5	756	$3 \times 3 \times 3$	$970 \times 856 \times 1109$	2.10GB
13P4	756	$3 \times 3 \times 3$	$963 \times 857 \times 1110$	2.10GB
17P8	756	$3 \times 3 \times 3$	$969 \times 849 \times 1117$	2.10GB
19P3	756	$3 \times 3 \times 3$	$918 \times 1147 \times 1011$	2.24GB
25P2	756	$3 \times 3 \times 3$	$1011 \times 921 \times 1153$	2.37GB
26P7	756	$3 \times 3 \times 3$	$955 \times 865 \times 1112$	2.01GB
30P1	756	$3 \times 3 \times 3$	$972 \times 1069 \times 1155$	2.74GB

We demonstrate our approach on the *AI Lattice* and test on real-world datasets to validate the end-to-end workflow, including comparing to ground-truth data from simulation (*Five Defects*), managing large-scale volumes (*AMTruss*), and processing experimental data (*2P10*, *5P6*, etc.)

laptops (see Section IX) we observe end-to-end runtimes well within practical limits (tens of minutes for small lattices; a few hours for our largest $9 \times 9 \times 9$ scan).

V. DOMAIN OBJECTIVES AND COLLABORATION

Over the past two years, our team, including material science and non-destructive evaluation experts, has engaged in regular project meetings. This close collaboration allowed us to observe the workflow and gain a nuanced understanding of key challenges in the domain. Following the suggested principles by Sedlmair et al. [48] we adhere to a iterative dynamic process when developing our visualization solution. Due to the close interaction with the project partners, some have become co-authors of this paper, having contributed significantly through regular brainstorming sessions, setting domain goals, generating data, and testing our approach. Because our organization's application area is highly specialized, we had to recruit users internally to evaluate our approach. We derived an empirically grounded understanding of the challenges and were able to quickly validate our assumptions. This approach also ensured that our contributions were directly applicable to the real-world challenges identified during our projects. The current workflow starts with the acquisition of the XCT scan, which is stored on a shared drive. Various team members and stakeholders access datasets by downloading them from their local machine for various tasks, such as developing analysis scripts and visually inspecting them slice-by-slice. The results are then uploaded to the same drive. This requirement to download entire volumes is a major obstacle for many stakeholders when trying to inspect the data. Furthermore, users often report resorting to shipping hard drives to collaborators due to the data size. Based on these challenges in the existing workflow, we derived the following goals:

- *Democratizing data access (G1)*: The traditional workflow involves multiple experts and organizations, creating barriers to data access and inspection. Despite using commercial products like VGStudio (Volume Graphics GmbH, Heidelberg, Germany) and Dragonfly (Nano Dimension,

¹ LLNL Open Data Initiative, UC San Diego Library Digital Collections, <https://library.ucsd.edu/dc/object/bb5504389k>

Waltham, MA) for manual inspection, various members, such as the material scientists, often lack direct access to the results, impeded by the data's size and complexity of interpreting volumetric data and lattice structures. Our first goal is to enable all involved experts to easily view and interact with the volumes.

- *Enabling quick overview of part quality (G2)*: The substantial size and complex internal geometry of XCT datasets hinder quick inspections. With thousands of internal struts even in small parts, slice-by-slice analysis is inefficient. Even volume rendering is limited due to occlusion and memory constraints. Thus, our goal is to let experts swiftly navigate and assess overall part quality from the XCT data.
- *Navigating to problematic struts efficiently (G3)*: Often only a small location of the part is problematic, but identifying that region is like finding a needle in a haystack according to our collaborators. One reason is that, once an XCT scan is performed, the inherent hierarchy of unit cells and struts in the part is not represented by the uniform grid of density values in the volume. Our goal is to utilize this inherent organization and enable intuitive navigation to and selection of problematic struts, facilitating the user to focus their attention only to relevant aspects in the data.
- *Relating defects to their spatial context (G4)*: Identifying defects in relation to strut orientation and location within the lattice is crucial for understanding their effect on mechanical properties. This goal focuses on enabling experts to contextualize defects within the lattice structure, aiding in understanding defect patterns and causes.
- *Streamlining analysis and validation of strut defects (G5)*: Whether defective or not, each strut can be characterized by various metrics, such as geometry, surface roughness, or breakage. The manual inspection of large numbers of these struts is not feasible for thousands of struts. Instead, we aim to develop dedicated metrics and visual representations that allow filtering and quick validation of strut condition, enhancing inspection efficiency.
- *Facilitating custom analysis and modularity (G6)*: Our collaborators have diverse use cases and continuously develop new test parts and defect characterization methods. Thus, it is essential to provide a modular framework that can be easily adapted to incorporate new analysis tools.

VI. VISUALIZATION DESIGN OF LATTICEANALYTICS

In response to the challenges in the current characterization workflow and goals described in Section V, we created the *LatticeAnalytics* framework which integrates a dashboard, a novel registration approach, a processing workflow, data infrastructure, and custom-designed visualizations to address the needs of large-scale XCT data management and effective metrics computations. To manage the large data from XCT scans, we implemented a network-attached storage (NAS) system where XCT data are converted, uploaded, and streamed (G1). Our web-based dashboard allows users to inspect the XCT data and the metrics seamlessly without the need for additional software installations, promoting rapid interaction and inspection (G1,

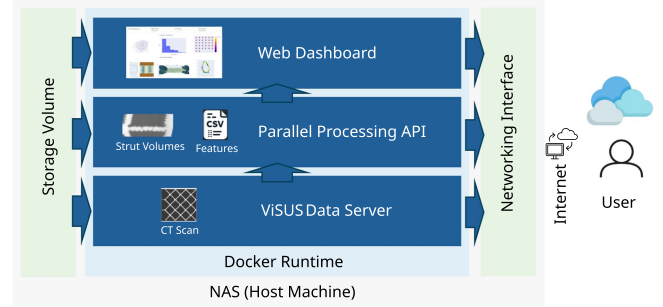


Fig. 5. We deploy our framework onto a NAS, where large-scale volumes are converted into a multi-resolution format (OpenVisus IDX format). We use a docker container for fast, hardware agnostic deployment. Clients and users can query sub-volumes or downsampled data or access the data via the dashboard or the API.

G2). The first step of our approach is to align the spatial graph with the CT scan, enabling selective processing of subspace queries for individual struts and facilitating parallel processing (G2-G6).

A. Large-Scale CT Data Management Infrastructure

Before considering the complexities of the data, the first problem is the dataset size. To enhance accessibility (G1) and facilitate analysis (G2-G5), we employ OpenVisus to convert the data into a multi-resolution format, allowing for rapid querying and downsampling of data as described in Section IV (see Section VI-A for details on the large-scale volumetric dataset format). Subspaces (e.g., slices or individual struts) are only a fraction of the volume in size and can be quickly queried over the network, especially at low resolution. Besides the location for this query, another parameter is the downsampling rate, to enable further data reduction and increase inspection speed. Storing data on the NAS and enabling multi-resolution queries greatly improves data accessibility as well as the efficiency of volume visualization. A schematic of the system is described in Fig. 5. To enable a simple deployment of our system, we use a Docker container that contains the dashboard and all processing algorithms.

B. Spatial Graph to CT Scan Registration

Data registration, nominal to actual comparison and consequently detecting deviations, is complicated by the noise inherent in XCT data (G3), as described in Section IV. In our case, we are not registering volume to a surface, such as a CAD model, but aligning to the nodes and edges of the lattice's spatial graph. Developing a fully automated algorithm is unfeasible, as there are defects in the parts, which naturally deviate from any nominal models. Prior works [30], [49] demonstrated the efficacy of using VR for higher complexity 3D models. We take inspiration from their success in lowering errors in dimensional estimates, and choose VR for the initial alignment task. This method reduces a difficult alignment problem to an expert-guided approach that consists of a coarse VR alignment and a fine registration step.

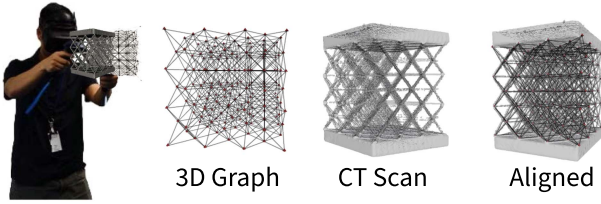


Fig. 6. We facilitate a coarse alignment using a VR environment that allows the user to deform the spatial graph to fit the CT scan. The resulting alignment will be fine-tuned in an offline registration algorithm.

1) *Coarse Alignment in VR*: Achieving 3D alignment on a 2D desktop is typically challenging due to limited depth perception and the unnatural mouse interactions with a 3D scene. We enable the user to load spatial graphs, either in a custom JSON format (see Section IV) or the Beam Lattice Extension portion of the 3MF format, which is commonly used in additive manufacturing to encode lattices. Our VR approach allows users to manipulate the spatial graph directly within the 3D volume using control points, thus improving alignment accuracy (Fig. 6). This interface uses a progressive refinement approach, mitigating precision issues from involuntary hand movements (G3), as proposed by Rodrigues et al. [26]. The alignment can be further adjusted by moving the control points individually while using a spring-model deformation to achieve a non-linear deformation of the spatial graph. The result of this step is that all nodes the spatial graph are close to but not exactly at their actual positions in the volume as a trade-off for a fast user-driven alignment. This alignment step in VR is the only manual intervention in our proposed processing workflow.

2) *Fine Registration of Nodes*: Following the manual coarse alignment of nodes, the next step is to move their positions to the node's center location. Each node's position is fine-tuned by analyzing a small window around the VR aligned node position. This location typically has the highest density after we apply a Gaussian smoothing window around this expected location. We identify the peak intensity as the node's center. We set the window width equal to the nominal strut length to avoid ambiguous cases where two nodes are contained in the window. This step is parallelized to speed up the registration process and ensure the spatial graph's accuracy relative to the XCT data (G4, G5). The difference between the coarse VR alignment and the fine registration is shown in Fig. 7, where it demonstrates how the manual imprecision gets remedied by our fine registration algorithm. Depending on the XCT resolution, this approach may still be susceptible to artifacts in the volume, which could position the node incorrectly. However, as we aim to filter defective struts, any such artifact would bias our metrics toward false positives – which the expert can quickly disregard using the dashboard.

C. Identifying and Fitting Struts

While experiments are initially conducted with small-scale test objects (Table I), the ultimate goal is to apply our methods to substantially larger XCT scans that can surpass several hundred

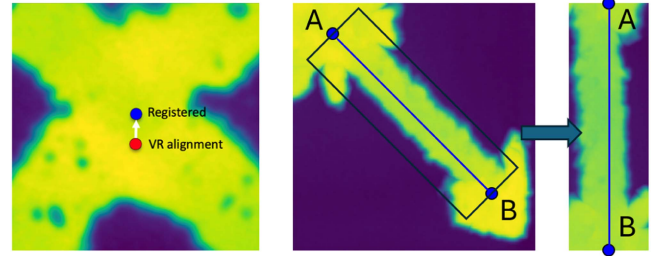


Fig. 7. The extraction and normalization of the strut and its orientation is shown in three steps here. (Left) Our algorithm takes the VR aligned node position and moves it toward the highest intensity values to create correspondence between the node positions of graph and the lattice in the volume. (Right) Using two adjacent nodes, we can then extract the sub volume around a strut and then rotate this sub volume to normalize the orientation for all struts.

gigabytes. Handling such large datasets presents significant challenges, not least due to the difficulty in loading entire volumes into memory and the sheer number of struts, which make manual inspections an extremely tedious task. To address these challenges, we have developed a novel metric computation approach utilizing the node-edge topology of lattice structures and pre-processed data in batches of sub volumes, each containing a single strut. Our collaborators provided metrics of interest that can be used for detailed characterization of defects, which can be computed on these sub volumes individually, allowing for parallelization of the processing workflow in a straightforward way. In the following we describe how these metrics can be computed on the single strut sub volumes in our approach.

1) *Sub Volume Extraction and Normalization*: Following the registration of the spatial graph with the CT volume, we identify the sub volume around each strut, which is defined as a rotated cuboid between the two registered and adjacent nodes, A and B (Fig. 7). First we define a cuboid that can fit a strut along its longitudinal axis and add a 20% margin to account for potential imaging and registration artifacts. Based on the orientation of the edge in the spatial graph, we rotate the sub volume to align with the z-axis. This normalization of the strut rotation simplifies the computation of any strut-specific metrics. This ensures consistent orientation where the first Z slice contains the node A and the last Z slice contains Node B at their center respectively. We store the coordinates and rotation of the strut to allow the dashboard to retrieve that sub-volume on demand. The biggest advantage of this approach is that struts can be queried on demand without ever loading the full resolution volume. The drawback is that the voxels associated with a node may be queried multiple times (proportional to the degree of edges), if adjacent struts are inspected. On the other hand, we skip a large part of the empty space between struts using this method.

2) *Segmenting Strut and Inclusions*: For segmentation, we implement a simple approach using Otsu thresholding, tailored to the homogeneous material properties expected from the uniform metal powder used in LPBF. The density histogram of the volume typically exhibits a bimodal distribution, reflecting the material and void spaces (air), making Otsu's method well-suited for differentiating between these groups of density values. The segmentation approach also facilitates the subsequent

TABLE II
A SET OF NUMERICAL METRICS THAT CAN BE DERIVED ON AN INDIVIDUAL STRUT BASIS

Metric	Computation
Length	The sum of euclidean distances between vertices of the centerline.
Mean/RMS Curvature	The mean or root mean squared of all angle deviations from straight line (180 degrees) at each point on the centerline.
Porosity	The number of inclusions detected by the segmentation algorithm.
Circularity	Dividing major and minor axes of the fitted ellipses.
Avg. Thickness	The average of the major axes of the fitted ellipses.
Volume	The voxel count of segmented strut.

identification of pores, such as trapped gas pockets and unmelted particles, by excluding the background and focusing on delineating between these two peaks using the same segmentation approach.

3) *Fitting Strut*: To determine key metrics such as strut curvature and trajectory, we first extract the center of the strut from the segmented volume. Volumetric skeletonization would not be suitable here due to potential distortions from fabrication and XCT scanning artifacts, which would result in a skeleton with substantial branching and interruptions. Given that struts are designed as straight elements with circular cross-sections, any deviations such as bending or breakage are significant. Our approach fits an ellipse to each cross-section (from the segmentation mask) of the strut that has the minimum distance to the cross-section's contour. The first advantage of parametrizing the contour with an ellipse is that it enables us to derive descriptors of the aspect ratio of the cross section. The second advantage is that the center of the ellipse can be considered to be the center position of the strut. The series of these center positions establishes a polyline representation of the strut's centerline. Laplacian smoothing is applied iteratively along the polyline to account for potential artifact-induced deviations. This strut parametrization as a polyline where each vertex is the center of the fitted ellipse along the strut volume allows the experts to quantify the deviations and provides a detailed characterization of each strut's actual trajectory compared to its nominal design, for example to understand the effect of gravity on bending of the strut.

D. Lattice and Strut Metrics Computation

For each sub-volume, its segmentation mask and the computed centerline, we demonstrate how our approach facilitates computing various numerical metrics suggested by our collaborators as a starting point for defect quantification (G4). Standardization of these metrics is an ongoing research topic, hence the metrics provided here are suggestions of how they may be computed; however, custom metrics can be computed by the experts through our API, as described in Section VI-E4. This functionality serves also directly G6, as we provide the flexibility to access the raw data and extend the predefined metrics to develop new calculations tailored to the scientists' specific research needs. An initial set of numerical metrics derived from the centerline and segmentation, calculated in physical units, are detailed in Table II.

E. Visualization of Strut Defects

A significant challenge in analyzing lattice structures is the overwhelming number of struts, each uniquely oriented and requiring inspection from multiple angles in 3D. While we can compute numerical metrics, they might not be reliable as their precision is dependent upon the resolution and presence of artifacts in the XCT scans. According to our domain experts, they need to validate these metrics and strut quality, however, inspecting all struts and their 3D geometry is rarely feasible. To address this challenge and to facilitate an efficient overview of potential strut defects (as outlined in G2 and G3), we developed two static 2D visualizations that allow for an unoccluded view of the two key metrics, strut geometry and roughness. As static overview images, the user can quickly browse through them without the need of user interaction. The first, the *Contour View* (CV), provides a rapid, 2D depiction of strut geometry, avoiding the complexities of 3D interaction. This view simplifies the initial assessment of strut shape and orientation, enabling quicker navigation to specific areas of interest within the volume. The second visualization, the *Roughness Map* (RM), focuses on the surface roughness of the strut, which is critical for understanding the lattice's fatigue resistance. Rather than relying solely on volume rendering or a single metric, the RM unwraps the strut along its longitudinal axis, presenting detailed, point-wise surface roughness in a 2D view.

As shown in Fig. 1, the views of the dashboard are linked between the top-row views, which focuses on the features of all struts, and the bottom-row views, which are centered on the individual strut. Furthermore, by hovering the mouse in the 3D iso contour view, the respective slice in CV and the z position in the RM are shown. Together, these visualizations form part of a comprehensive dashboard that includes multiple coordinated views.

1) *Contour View*: The CV offers a novel 2D representation of strut geometry by projecting all contours along the longitudinal axis from the normalized segmentation mask onto a plane. As outlined in Section VI-D, for each slice along the longitudinal axis, we derive the contour from the segmentation mask. We use the same window size for all struts in the structure. This way, we ensure that *abnormal* sized or positioned struts are obvious in the visualization. These contours are color-coded based on their z-position using a sequential color map; slices closer to the viewer are represented in brighter colors, while those further away are shown in darker colors. This color gradient helps in visualizing depth in the 2D plane.

Next, the color-coded contours are projected along the z-axis and blended in a first-hit manner where the nearest (brightest) contours are rendered and occlude the farther contours, creating a ordering by depth. This method is depicted in Fig. 8. The primary advantage of this view is that it allows for a rapid assessment of the strut's geometry and any defects without 3D interactions while only marginal 3D information is abstracted. As shown in Fig. 9, properties such as protrusions, inclusions, the axis ratio of the cross-section, and the overall shape are easily identifiable in this standardized 2D view. Struts that are off-center or defective are immediately apparent in this view,

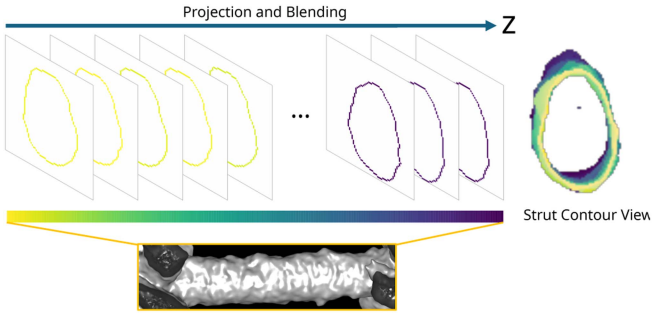


Fig. 8. *Contour View*: We color each contour along the z-direction of the segmented and normalized strut volume. All contours are projected onto a 2D plane.

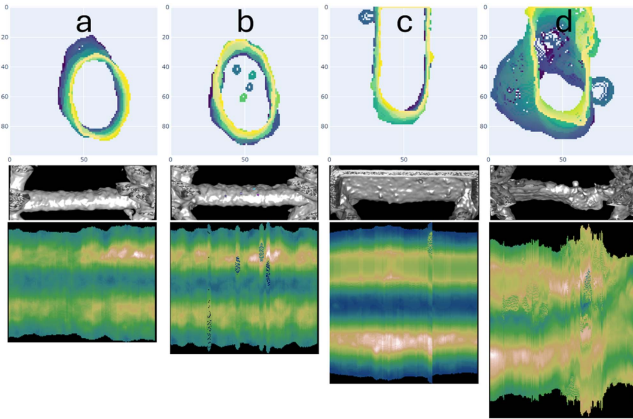


Fig. 9. Examples of *Contour Views* (CVs) and *Roughness Maps* (RMs) from the A1 Lattice: This figure showcases various strut analyses where the CVs and the RMs allow for fast visual assessment of different type of defects. The different width of the RMs originate from varying circumferences between struts. (a) A normal strut with an aspect ratio of 1.48, as depicted in the CV. The RM highlights areas of surface roughness on the top side. (b) The CV reveals four inclusions and a uniformly coarse top side, alongside a fairly even overall shape. Correspondingly, the RM shows these inclusions and roughness on the top side. (c) This CV shows that the strut is fused with the slide top of the object, visually indicated by the cut-off in the top of the CV. The RM also reflects this shape, with additional elevations on the bottom side. (d) A severely bent strut featuring multiple surface defects. In the CV struts is partially cut off as it wanders out of the window of the plot. The particularly rough surface is also indicated by the RM.

allowing the user to recognize these issues at a glance. While the CV efficiently highlights general shape characteristics and potential bending in the struts, some issues like breaks may be less visible due to occlusion by nearer contours. However, in our interface, we provide a complementary full-resolution view of the strut volume for detailed inspections, as discussed in Section VI-E3.

2) *Roughness Map*: Assessing strut roughness visually within a volume rendering often necessitates interaction with 3D visualizations from multiple angles, complicating fast or comparative analysis across struts. This method also tends to be subjective. To overcome these challenges while providing a quantifiable and easily comparable roughness analysis, we developed the *Roughness Map*. This visualization technique leverages the cylindrical nature of struts, unwrapping them into a 2D plane along their longitudinal axis, thereby avoiding the

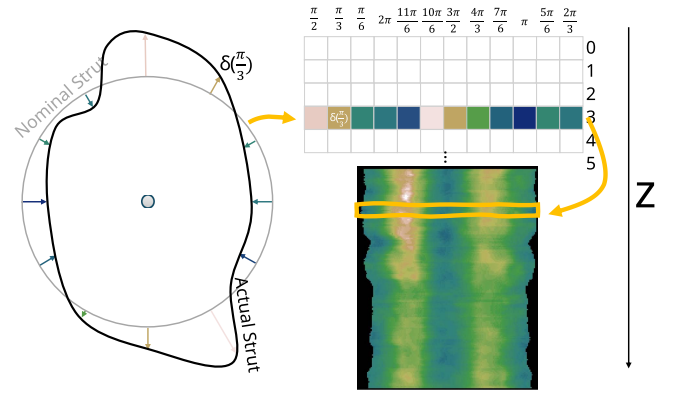


Fig. 10. Illustration of *Roughness Map* approach for a single cross section. We map the distance between the nominal contour (circular) to the actual contour for each cross section along the strut. We compute the polar coordinates and the difference (δ) between the actual radii and the nominal radii is mapped to a diverging color map. We store the δ values as a row in matrix where each row index corresponds to the z slice of the contour. This view reveals protrusions, cavities and general roughness of the strut.

complexities of 3D navigation. As depicted in Fig. 10, our algorithm computes local roughness by calculating the radial distance in polar coordinates from each contour point to its center. These distances (δ values) are stored in a 2D matrix in clock-wise fashion while starting at the $\frac{\pi}{2}$ position for all contours. Hence, each row corresponds to a contour and depicts the point-wise distance from the center. This matrix representation preserves the spatial relationship of roughness along the strut's surface. To enhance the visual interpretation of these roughness measurements, we use the metaphor of a topographical map. Elevations on this map indicate structures protruding from the strut surface, such as semi-molten material, while depressions highlight cavities or pores resulting from incomplete melting or other process instabilities. We utilize a diverging color map typical of topographical maps, where the nominal radius ($\delta = 0$) of the strut cross-section serves as the “sea-level”.

This visualization not only facilitates a direct visual assessment of surface irregularities but also highlights differences in cross-sectional shape. For example, in Fig. 9, two vertical smears indicate an elliptical rather than circular cross-sectional shape, with generally longer radii at the top and bottom. The RM immediately reveals areas where material protrudes or recedes, providing a clear, intuitive understanding of strut surface roughness.

3) *Visualization Dashboard*: The dashboard is the visual interface of our framework, providing web-based access to both extracted metrics and raw data for all organizational stakeholders. Opting for a web-based interface, we reduce the need for installing standalone packages and ease the access to our new capabilities (G1) for domain experts.

a) *Dashboard Structure and Functionality*: The dashboard is designed with custom views that allow users to quickly select specific struts based on their metrics or dive deeper into a detailed inspection of individual struts. There are three sections in the dashboard: the control bar, the first row views showing metrics and allow for selection, and the second row views

showing the strut-level visualizations of the selected strut. The interface elements (upper case letters) refer to Fig. 1 and function as follows:

- *Control bar*: The control allows the user to select the metric of interest and adjust the resolution of the query (A). Considering that even sub-volumes may be too large for download over the network, the dashboard allows users to subsample data directly in the query (*Strut 3D Quality Slider*), facilitating faster viewing. This feature is essential for validating computed metrics against the raw data, ensuring that analysis is grounded in accurate representations (G6).
- *First row views*: The selected metric will be visualized in a Histogram View (C) for all struts, enabling users to see the distribution and select a specific range or bin, typically finding abnormal struts on the tail ends. This selection process directly supports the goal of rapid assessment and comparison across struts (G3). This interactive histogram widget allows the user to zoom into the histogram for a precise selection of bins. The struts in the selected bin are then shown in the Spatial Graph View (B), highlighted in green. This visualization aids in identifying spatial and orientation-based trends within the lattice, aligning with goals G3 and G4. The Detailed Strut Analysis View (D) visualizes the feature per-strut as a colored circle, which also facilitates the selection of individual struts for further analysis (G2, G3). The selected strut is then highlighted in Red in the Spatial Graph View and the linked views in the second row will show the visualizations for the selected strut.
- *Second row view: Contour View (E) and Roughness Map (G)*: Offer quick, intuitive assessments of the selected strut's quality, complementing each other by providing different metrics on strut geometry and surface characteristics as described in Section VI-E (G2). *Multi-resolution Volume Rendering (F)*: For a more detailed inspection, the selected strut volume is visualized using iso-surface rendering. The centerline as well as pores are overlaid. This view is crucial for observing the strut's actual shape as well as internal features and is generated from a rotated bounding box query from the whole volume as described in Section VI-A (G4, G5).

4) *Providing API Access for Custom Computations*: To accommodate the diverse needs and use cases of our expert users and enhance the flexibility of our system, we provide a Python API that allows for the customization of data analysis workflows (G6). This API facilitates access to the raw data and the computed metrics.

Custom Analysis of Existing Metrics: Metrics for each strut are stored in tables, as detailed in Section IV. This setup not only allows users to retrieve and analyze these metrics but also supports the integration of this data with external tools for advanced statistical analysis or visualization. For instance, users could generate a scatter plot comparing aspect ratios across strut contours to study geometrical variations within the lattice, enhancing understanding of structural integrity.

Development of New Metrics: Beyond analyzing existing metrics, users are able to define and compute new metrics using the established registration with the spatial graph or the extracted strut sub-volumes. This capability is crucial for tailoring the analysis to specific requirements, such as, implementing custom algorithms for segmentation or refining the extraction of centerlines based on unique project needs.

Efficient Data Querying and Visualization: Our API also supports querying data from the overall volume at various resolutions or specific locations, which is particularly useful for handling large datasets efficiently. Through our integration with the OpenVisus server, users can access multi-resolution data, allowing for flexible data visualization and manipulation. For example, during the VR alignment process described in Section VI-B, users can load a lower resolution version of the volume to facilitate quicker and more efficient alignment adjustments.

VII. IMPLEMENTATION AND CONTAINERIZED DEPLOYMENT

We developed the web-based dashboard using Python, which leverages several advanced libraries to handle the streaming and processing of high-resolution XCT data efficiently. The core technologies include the following packages. *Plotly and Plotly Dash* (Plotly Technologies Inc, Montreal, Canada) are utilized for creating interactive dashboards and visualizations. *OpenVisus* is employed for converting and streaming multi-resolution volumetric data. *Pandas*, a software tool for Python, is used for managing metadata associated with the lattice as well as the tables described in Section IV. As depicted in Fig. 1, the dashboard features a user interface with six main plots organized in a 2x3 grid. The image-based algorithms are implemented using *pyblesperanto*, a Python package. These algorithms take advantage of GPU acceleration and parallelize over the strut sub volumes. The VR-alignment was developed using the Unity Game Engine (Unity Technologies, San Francisco, CA), and we tested it with several VR headsets, including HTC Vive, HTC XR Elite, and Meta Quest 3, facilitating intuitive, real-time spatial alignments. This workflow is designed to be deployable across various systems, including personal devices, NAS, and high-performance computing platforms. The initial deployment was carried out on the NAS, supplemented by desktop computers equipped with GPUs for intensive processing tasks. To ensure security and practicality, we encapsulated the dashboard and processing workflow within a container, deployable via *Docker* (Docker, Inc., Palo Alto, CA) or *Podman* container engine. For our use case, this containerization strategy provides several advantages (G1):

- *Isolation*: Keeps the application isolated from the host's operating system, minimizing potential security risks.
- *Minimal privileges*: Operates with the least privileges necessary, enhancing security.
- *Environment consistency*: Guarantees that the application runs consistently across different computing environments without the need for manual installation of dependencies.

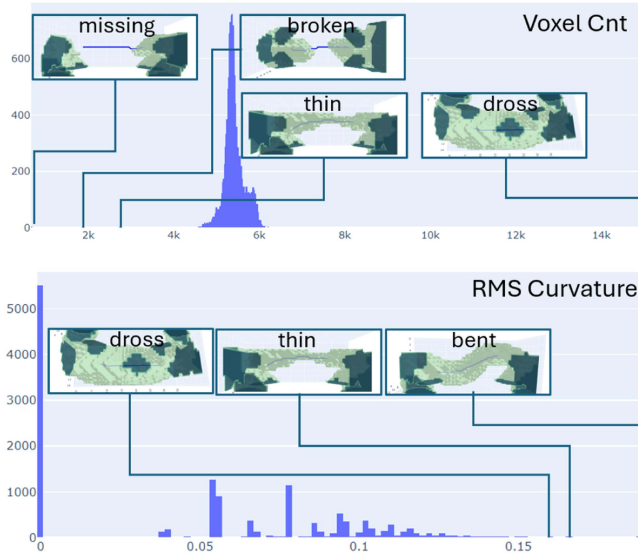


Fig. 11. The histogram of the voxel count and the RMS curvature reveals all five defective struts as outliers, which can be easily selected in our dashboard.

VIII. RESULTS AND CASE STUDIES

Our methodology is designed to be universally applicable to any XCT scans of strut-based lattice structures, whether they are regular lattices, meaning following the octet truss topology, or irregular topologically optimized ones, provided that a spatial graph is available for our registration approach.

A. Revealing and Visualizing Lattices With Ground-Truth Defects

We demonstrate our approach on a publicly available dataset our team previously published [27], as described in Section IV. This simulated XCT scan of an $8 \times 8 \times 8$ octet lattice structure has a total of 13,056 struts where five struts have typical geometrical strut defects (*dross*, *bent*, *thin*, *missing*, *broken*) at known-locations. This ground-truth data allows us to evaluate how our approach helps finding these suspicious struts without having to browse through the entire dataset. We first inspect the voxel count as an approximation for the volume of a strut. The voxel count histogram, shown in Fig. 11, reveals four single-element bins on the left and right tail ends. This metric alone already reveals the *missing*, *broken*, *thin*, and *dross* defect, as they typically deviate from the *normal* struts in this metric. Furthermore, we also view the histogram of the *RMS Curvature*, which shows three single-element bins at the right tail-end. By selecting the bins, these turn out to be the struts with the *dross*, *thin*, and *bent* defect. As these are purely geometrical defects, we can also utilize our CVs to view struts without viewing the 3D volume. Fig. 12 demonstrates that the CV of the five defective struts look substantially different than a normal one. This use case shows how quickly our dashboard reveals all five defects as outliers in specific metrics and confirms that the CV of defective strut appear different from normal ones.

Instead of utilizing the available metrics through the dashboard, we can also derive custom metrics using the provided



Fig. 12. While the normal strut appears almost as overlapping circles, the defective ones show significant differences. For missing and broken struts, part of the missing color contours indicate that strut is incomplete. Furthermore, the smaller contours towards the center indicate that the strut is thinning towards the center. The CV of the bent strut clearly shows the change in the trajectory. The dross defect appear as widening contours due to the additional material buildup. The CV of the thin strut shows that the contour are increasing in size but shows all contours, indicating there is no breakage.

API to the dashboard. We showcase a Python function that extracts all *broken* struts. This function first queries the strut and creates the density profile by sampling along the centerline of the queried strut volume. As shown in the below sample code, the function takes one argument, the *relative_broken_thresh* to define the percentage in the density profile that fall below the otsu threshold used in the segmentation in order for the strut to be classified as *broken*. This function correctly returns both the broken and missing strut.

```
1 def get_broken_struts(strut_ids,
2   relative_broken_thresh):
3   broken_strut_ids = []
4   for strut_id in strut_ids:
5       strut_vol = query_strut(strut_id, 0)
6       centerline_density_profile = [sample_strut(
7           c_pos, strut_vol) for c_pos in
8           get_centerline(strut_id)]
9       if ((np.sum(centerline_density_profile <
10          get_material_thresh(strut_id)) /
11          get_length(strut_id)) >
12          relative_broken_thresh):
13           broken_strut_ids.append(strut_ids)
14   return broken_strut_ids
```

B. Inspection of Strut Metrics Independent of Data Size

Currently handling large-scale XCT data presents a dilemma for our collaborators: such datasets are infrequently acquired due to the challenges in managing and processing them, as discussed in Section V. Consequently, developments in this area are limited due to the rarity of these large-scale volumes. Most datasets our collaborators provided were of a scale manageable on standard desktop machines. However, to validate our approach's effectiveness irrespective of data size, we applied it to the AMTruss dataset, which is known for posing significant challenges in current workflows due to its large size.

As illustrated in Fig. 13, our method effectively filters struts based on their curvature to identify bent struts. The 3D lattice view offers a comprehensive overview of the locations of these bent struts, while the iso-surface rendering and the CV highlight significant variations in strut thickness around their midpoints, indicative of curvature changes. Despite the dataset's substantial size (91.33 GB), our approach facilitates real-time inspection of bent struts without the need to download the entire volume. This

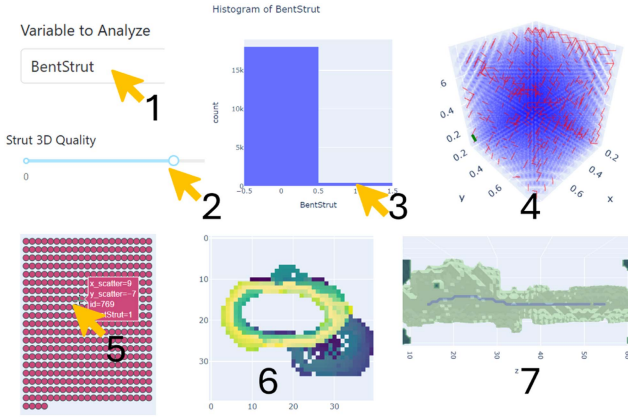


Fig. 13. We demonstrate how to operate the dashboard when applied to the large-scale AMTruss volume (91.33 GB) where we show the identification of a bent strut. 1) The user starts by selecting the bent strut metric and 2) adjust the resolution for the volume query. 3) Since this is a classification, there are only bent and normal struts in the histogram. By selecting the positive bin. 4) The selected bins will then be shown in the spatial view to reveal their overall location and orientation. 5) The user selects one strut. The circles (representing the struts) are all the same color as they depict only one category. 6) The CV reveals that the struts has a bulge on side. 7) After closer inspection, it can be confirmed that this is indeed a bent strut. The analysis is facilitated within seconds, depending on networks speed. Only the pre-computed metrics are transferred to the client.

capability significantly accelerates the process, allowing for a rapid and detailed quality assessment of the part.

C. Automatizing Manual Quantification

Our collaborators provided us with ten datasets to further validate the efficiency of our system on data that caused significant challenges. They consist of ten $3 \times 3 \times 3$ lattices, detailed in Table I. These lattices were specifically printed by our collaborators to refine defect characterization techniques. According to our collaborators, tasks including segmenting struts and nodes, calculating their average sizes, and identifying broken struts required approximately one month of manual effort by experts in non-destructive testing and material sciences. In contrast, our automated approach processed all ten datasets within a single day using a GPU-equipped laptop (RAZER BLADE 16). This process required minimal manual intervention – only the initial spatial graph alignment, which a team member performed in VR in under a minute per dataset. The processed results were promptly made available through our web-based dashboard, enabling experts across the organization to easily access, visualize the geometry and surface roughness in the CVs and RMs, and inspect these datasets. The effectiveness of this approach and the feedback received from users are further discussed in Section IX.

IX. USER EVALUATION

This work was reviewed and approved by Lawrence Livermore National Laboratory (LLNL) review committee (LLNL-JRNL-871130). All material and methods were carried out in accordance with relevant guidelines and regulations. To evaluate the usability and usefulness of our proposed system, we performed exploratory and semi-structured interviews with domain experts in our organization. In the following sections,

we describe the characteristics of participants, the design and procedures, and qualitative feedback. The datasets from Table I were used in this study.

A. Participants

We conducted the exploratory and usability testing as well as semi-structured interviews with five domain experts (E1 – E5). Three of them were males and two were females. Three participants (E1, E3, and E4) rated as having working experience between 6 to 10 years, E2 rated between 3 to 5 years, and E5 was in a senior role and had over 10 years of working experience. Their participation was voluntary. They were all material scientists specializing in the additive manufacturing research and the characterization of AM parts.

B. Apparatus

We used a laptop RAZER BLADE 16 for demonstrating the VR alignment and visualization interface. The laptop was equipped with a 13th Gen Intel Core i9-13950HX processor with 32 cores, an NVIDIA GeForce RTX 4090 graphics card with 16 GB of VRAM, and 32 GB of RAM. During the VR testing, we used a Meta Quest 3 headset with a resolution of 2064×2208 pixels per eye, a 110-degree FOV, and a refresh rate of up to 120Hz.

C. Study Design and Procedures

We conducted the guided exploration and semi-structured interviews to achieve two main research objectives. First, we aimed to elicit the experts' informed opinion regarding usability, potential benefits of the tool, and its applicability to their workflow. The second objective was to obtain feedback and suggestions for further improvement. Before starting the study, all participants were informed about the objectives and procedure of the study, and their verbal informed consents were collected. During the study, two researchers were present with the participants. The first researcher introduced and explained the system features as well as assisted the participants to explore the system. The second researcher assisted with the VR demonstrations and took notes during the discussion and interviews.

The procedure starts with introduction and research objective explanation. The participants were informed about procedures and collected data. Their verbal consents were collected. They were also asked to fill out the demographic questionnaire, including their expertise and working experience. The procedure was divided into two phases: (1) exploration of VR alignment and visualization interface, and (2) semi-structured interview. After the completion of the exploration, participants were prompted to share their overall impression and feedback on the usability, benefits, applicability, and limitations to further improve the system. Finally, they were asked to complete the user experience questionnaire.

D. Data Recording and Analysis

We followed a think-aloud protocol to gather qualitative data during the exploration. For semi-structured interviews, a

researcher took notes of all relevant comments and the conversations were recorded. Those comments were collected and stored in a common database. They were then categorized into relevant topics and research objectives. The usability and usefulness were assessed with a 7-point Likert scale from “strongly disagree” to “strongly agree” using module I of *meCUE* questionnaire [50]. We also collected an overall evaluation rating using Module V of *meCUE*, which ranges from -5 “as bad” to 5 “as good” with a scale interval of 0.5.

In the following sections, we describe results obtained from semi-structured interviews and qualitative feedback collected using the think-aloud protocol as well as the questionnaire. The average time of the study took approximately one hour, including the completion of questionnaires.

E. Expert Feedback

Overall, the expert feedback was positive. We also asked them about their current workflows, to confirm the challenges and needs (see Section V) were accurately identified. They appreciated the work and expressed interest in its potential benefits. The collected comments were assigned to four main categories listed in the following subsections.

1) *Current Approaches for Lattice Inspection:* Inspecting AM lattice structures is a crucial step for ensuring quality and performance, as well as informing design improvements. Experts described the tools they currently use for inspection. In addition to visual examination and XCT scanning, they use optical microscopes, optical surface profilometers, electron microscopes, structured-light 3D scanners, and even calipers for strut measurements. E2 and E5 mentioned software tools, including *Dragonfly*, *Avizo* (Thermo Fisher Scientific, Waltham, MA), and *Geomagic* (3D Systems, Rock Hill, SC) for inspection tasks. Besides these inspection tools, E1 stated that they still perform some steps manually – for example, aligning a CAD model to XCT data using VGStudio. E1 mentioned this registration was very time-consuming, especially on large-scale data. E2 echoed these issues, citing the high computational expense of current tools on their available workstations.

2) *Data Management and Parallel Computation of Metrics:* All experts agreed that a major challenge is handling the large-scale data from complex lattice structures. In particular, the size and volume of data continue to pose challenges in managing and transferring it between their teams. E3 described challenges in applying semi-automated, machine-learning approaches on their data – currently, they would run algorithms overnight due to the processing time. E4 revealed that they resorted to using *Microsoft PowerPoint* slides to present and discuss scan data, because creating static images was faster than trying to interact with the data in real-time. In addition to data management issues, E1 mentioned that flexibility in how they characterize lattice structures is another challenge in their workflow.

All of them agreed that the dashboard’s multi-resolution data streaming and querying features would help in managing and analyzing large datasets. E1, E3, and E4 emphasized that it is not necessary to visualize the entire volume; however, having

a tool to detect the differences (e.g., missing or broken struts) and visualize them in a comprehensive way would be essential. E3 remarked that while they generate a lot of CT data and can segment it, they lack the ability to view or analyze it quickly. Thus, the idea of breaking the data into smaller component parts (per strut) was very interesting and useful to them. E4 said, “*I like that a lot! When we first started taking a bunch of CT scans, we had people writing algorithms to analyze everything. This (dashboard) is what I had in mind – a way to visualize which strut is where, and which ones are broken, at a glance*”. E4 added that to get the most thorough analysis they always rely on XCT scans; however, without a tool like ours, they have to examine the data manually – measuring struts and flipping through cross-sectional slices.

3) Visualizations and Dashboard:

a) *VR Alignment:* E1, E2, and E4 expressed that the VR tool was easy to use, intuitive, and useful to perform the alignment. E3 stated that it took a bit of time to learn how to use it. Once they understood the user interface, however, they agreed it is significantly faster compared to desktop-based tools using mouse and keyboard. E4 agreed that users would need some initial VR training (e.g., how to use the controllers), noting that it’s a learning curve much like mastering shortcuts in a CAD program. E5 described the tedious process of alignment in a typical desktop 3D viewer (*Avizo*, in their case), and noted that aligning CAD/STL models with CT scans is a major challenge in general. Hence, performing the alignment in VR could provide a coarse alignment without having to manually align on a desktop or developing an algorithm. E5 – “*You could then align that subset of the 3MF [3D manufacturing format] to the right spot in the lattice that you care about and thereby rather than having the algorithm try to find all these [alignments], you can just essentially align them in VR*”.

E3 commented on how the VR alignment would have an impact while it is being used to get just approximate locations of the model. Within the immersive VR environment and natural interactions, the user can perform a coarse alignment quickly less than five minutes for each junction. Therefore, this coarse alignment allows the system to query the volumetric data and find the actual junction positions. The system will then perform the fitted alignment using Gaussian smoothing and perform other tasks for analysis, characterization, and visualization. E2 described another potential application in AM, which this VR alignment can be useful. During the printing process, they capture multimodal data, including printing path, machine data, and in-process images by setting up a camera to capture them layer by layer. It would be beneficial to use the alignment to make a comparison or image correlation, for example, after the mechanical testing. Hence, they could assess the failure and their deformations.

b) *Overall Dashboard:* All experts found the dashboard’s views are useful for providing a quick overview of the analysis. They specifically noted that the 3D lattice graph (showing the entire part) was helpful for visualizing overall structure and part integrity, as well as giving context such as strut dimensions and connectivity. The potential broken or missing struts are visualized in red, and it allows the user to select those specific

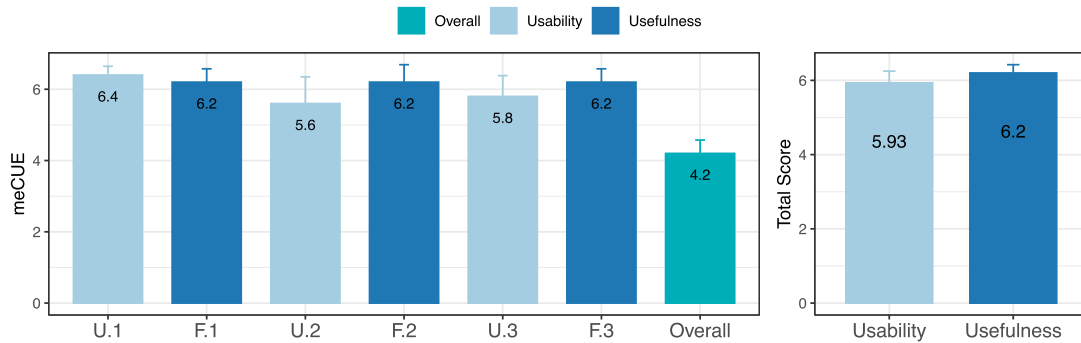


Fig. 14. Results of usability (U) and usefulness (F) as well as overall experience (Overall), including results of each item, using the *Module 1* of meCUE questionnaire (left) and their total scores (right). The rating scale of usability and usefulness ranges from 1 (strongly disagree) to 7 (strongly agree). The score for the overall rating was rated in a range between -5 (as bad) to 5 (as good).

struts for further analysis. E1 mentioned that it was easy to use and useful for machine operators, e.g., what to learn if the parts are good or broken and get a summary of the struts. E2 said *“I think that would be incredibly useful because from my understanding, you can check and get the way it’s done. Generally, it can load the entire dataset and then you can go in and manually pick out all these things.”*. E2 noted that even though the workflow was new to them, it was simple to grasp: select a strut in the skeleton plot, identify broken struts from the histogram, then inspect a chosen strut in detail via the 3D view, Roughness Map, and Contour View. E3 proposed a feature to link the orientation of the selected strut or node to different features, e.g., inclusion or surface roughness. Thus, it could provide a quick understanding of the strut’s orientation in other plots. Additionally, E2 encouraged exploring alternative color mappings to support the visuals – for example, using green to indicate good struts and shifting toward red for struts that are more compressed or have other issues.

c) Roughness Map: E2 noted that, to their knowledge, no existing tool offers per-strut surface roughness metrics or correlates them with structure. The experts were very interested in this capability, as it would let them evaluate how smooth a strut is, how accurate (in terms of geometry) it is, and how many potential stress concentration sites exist on each strut. E3 emphasized that the strut size, their shapes, orientations, and indicators for the broken struts are useful to visualize in the RM. Hence, it can be linked and correlated well with other features of the plots and their corresponding structures. E4 stated that surface roughness of struts is a factor they care about. They envision this RM feature will be even more important for non-lattice internal structures (such as gyroids or other shapes), where assessing surface area/roughness is critical. E4 suggested exploring alternative colormaps for the RM in the future, as different schemes might better highlight depth variations – similar to a terrain elevation map.

d) 3D Multi-resolution Strut View: For 3D strut volume rendering, E1 commented that 3D strut volume rendering was very helpful in providing a detailed view of the selected strut to validate the selected features. E1 noted that viewing the entire volume at full resolution is unnecessary; instead, they need a

tool capable of quick visualization with appropriate downsampling. Thus, our approach of computing per-strut metrics (and correlating them with visuals) aligns exactly with their needs. E2 said they would use this capability frequently, since it provides a quick way to visualize an individual strut instead of having to load the whole volume.

e) Contour View: Experts found the CV helpful for visualizing complex strut geometries in 2D by plotting constant Z-axis slices. It allows users to extract insights from intricate patterns and their relationship to immediately see geometrical anomalies and uncover underlying structures. E1 stated that this plot was easy to use and understand; however, they are not sure how to extract useful information from it without the complementary 3D view. On the other hand, E2 found that it is useful as an additional metric to characterize the lattice structure. E2 compared it to their current workflow of examining XCT scans slice-by-slice – noting that it’s challenging to extract each strut’s shape from the whole volume. The CV, by contrast, isolates each strut in a single view.

4) General Feedback: E1 suggested adding an indicator of the overall part quality – for example, displaying the total count of missing or broken struts on the histogram. They felt this would be essential for quickly evaluating if a part has an unusually high number of defects. They also suggested adding a heatmap view of the lattice, to highlight areas or regions that need more attention (e.g., with high defect concentrations). E2 emphasized that a colleague had spent around 10 months processing similar data manually, and said that if they’d known about this system – “it could have saved us like nine months”. E3 similarly remarked that features like per-strut surface roughness characterization were new to them – and because they hadn’t heard of such approaches. They found the ability to correlate multiple features (e.g., roughness, curvature, strut location) very useful. E3 thought it would be beneficial to include some statistical analysis of strut accuracy or defect levels at different locations in the lattice (e.g., how certain regions fare in terms of defect metrics). E3 also suggested an export feature (e.g., to a spreadsheet) so they could perform further analysis on the metrics outside the tool. Similarly, E4 and E5 suggested to add new features in addition to the visualization. Those include threshold adjustment

and filtering to see how many struts were broken or how many struts were below a certain threshold, average diameters or histogram, and export function as well as deploying source code for further analyze with their machine learning approach.

F. Questionnaire Results

Descriptive results of questionnaires for each item are listed in Fig. 14. The rating scale for usability and usefulness was based on a 7-point Likert scale, while the rating for overall score was in a range between -5 to 5. The total average for usability (U) was $M = 5.93$, $SD = 1.04$ and usefulness (F) was $M = 6.20$, $SD = 0.84$. The average score for overall experience was $M = 4.2$, $SD = 0.84$.

Trends in descriptive results, the usability can be classified as rather easy to use. *U.1* received the highest score among other questions ($M = 6.4$, $SD = 0.55$). For *U.2*, it could show that the user may need to get a quick introduction or training on how to use the system; however, it is a common procedure when using it for the first time. The operating procedure (*U.3*) can be assessed as fairly simple to understand. All experts found that the proposed system is useful and meets their expectations in achieving their goals.

X. DISCUSSION AND LIMITATIONS

This work describes a comprehensive pipeline designed to transition experts from a manual, tedious workflow to an innovative, VR-supported and visualization-enhanced process. Our approach addresses the challenges associated with accessing large-scale XCT scans in environments where design, fabrication, and inspection are carried out by separate teams. Prior studies have primarily focused on defect quantification to correlate mechanical properties. However, they have not addressed the challenges associated with keeping the expert in the loop, including data access, large-scale data management, metric extraction, and effective visualization.

Our framework significantly accelerates these defect characterization and research on lattice structures by improving data accessibility and introducing custom visualizations that allow for rapid quality assessments of part builds. The focus of this research was to develop a practical, end-to-end solution rather than targeting isolated aspects of the problem. Unlike prior solutions that address isolated steps, such as manual measurements (e.g., analysis in VG Studio or Dragonfly), extraction of global metrics (Klingaa et al. [4] and Yang et al. [7]), and utilize nominal to actual alignment (Klacansky et al. [27]), our end-to-end, human-in-the-loop framework offers integrated advantages. LatticeAnalytics reduces data handling overhead and targets defect detection locally, in contrast to one-size-fits-all full-volume analyses and proposes custom visualization for common strut defects. This yields a more efficient inspection workflow and allows experts to combine intuition with automation. Furthermore, the data management and fast remote access to the data via the dashboard promises to reduce the characterization task in geographically separated teams.

These visualization approaches proved extremely useful during evaluations, providing quick insights into key defect indicators. However, there are limitations at the current stage. The CV may suffer from occlusions where a closer contour occludes a farther contour, and the RM can suffer from distortions, typical to projection-based techniques. We accept these as necessary trade-offs to minimize complex 3D interactions and streamline the inspection process across different struts. At the current stage, our system is tailored for strut-based lattice structures commonly used in LPBF. While it does not yet support other internal structures such as gyroids, our project plans to adapt and expand our methodology to include these in future updates. A majority of the proposed pipeline can be adapted to these structures.

In addition to the case studies, we also performed a user evaluation through exploratory and usability testing. Our user evaluation was conducted with domain experts working with material science and additive manufacturing research. We followed a think-aloud protocol and conducted semi-structured interviews to collect qualitative feedback. This provided valuable insights on the system's potential benefits, usability, applicability in their workflow, and current limitations. The feedback was overall positive, and results of usability questionnaire could be rated as easy to use. However, the sample size was relatively small and based on subjective measures. Future work should aim to conduct an extensive evaluation, including objective performance metrics (e.g., task completion time, alignment error rates, user task success) as well as additional subjective metrics (e.g., cognitive load assessments and more detailed user experience ratings).

A. Implications

Results of the case studies and user evaluation as well as lesson learned and findings from our study reveal the following implications:

- Separation of the volume into separate chunks, each containing an individual strut, provides significantly advantages for revealing and visualizing strut defects.
- The proposed system provides a better solution for subsequent processing, since users can focus on local sub-parts of the data instead of applying a one-size-fits-all analysis to the entire dataset.
- Feedback from the experts revealed insights for each feature ranging from VR-based alignment and visual interfaces to general feedback, which could provide future actionable design recommendations.

XI. CONCLUSION

The research presented in this work supports materials scientists in characterizing defects in strut-based lattice structures – a capability crucial for advancing manufacturing in aerospace, medicine, and defense. This project tackled significant challenges often overlooked by the materials science community, including data access, effective visualization, and rapid quality assessment. Our flexible approach allows parts of our workflow to be adapted for similar applications beyond metal AM.

While our collaborating experts are specialized in metal AM, our technique is agnostic to the actual AM method, but could be applied to any lattice-based infills. Furthermore, our data management system could be used in a wide range of application that rely on XCT scanning. Beyond AM, our visualizations could be applied to similar tubular structures, such as blood vessel imaging. In particular, the CV could be adapted to show changes in large vascular structures, while the RM might be useful for depicting plaque buildup in blood vessels, provided the vessel trajectories are straightened beforehand. This framework has garnered continued support and funding, reflecting its potential and the recognition of its value by the material science community. Future enhancements will extend to include other internal structures, such as periodic minimal surfaces, broadening the scope and impact of our work.

ACKNOWLEDGMENT

We thank the experts Dr. Dino Wu (LLNL) and Jenny Nicolino (LLNL) for their valuable feedback on our approach.

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